

Oobleck

The Solid-Liquid Mystery

Session 1: The Rule Breaker Fluid

Magic Mud or Science?

Vivaan and Diya found a bowl of strange "magic mud" in their backyard. When they tried to punch it, it felt hard like a rock! But when they let it sit in their hands, it dripped away like water. Is it a solid? Is it a liquid? Let's solve the mystery today! [cite: 1]

Your Science Lab Goals

Today, we will become "Mad Scientists" to achieve these goals:

- **Understand Matter:** Review the basic rules of solids and liquids. [cite: 1]
- **Meet the Rule Breaker:** Discover why Oobleck is a "Non-Newtonian" fluid. [cite: 1]
- **The Pressure Test:** Learn how pressure changes the state of Oobleck. [cite: 1]
- **Make Your Own Oobleck:** Use simple kitchen ingredients to create magic mud! [cite: 1]

THE BIG QUESTION

If you could walk on a pool of liquid without sinking, would you try it? Oobleck makes that possible! [cite: 1]

Solids and Liquids: The Normal Rules

Most things in our world follow simple rules. Let's look at what we see in our Indian homes: [cite: 1]

1. Solids (The Firm Ones)

Solids hold their shape no matter what. A **Stainless Steel Glass** is a solid. It doesn't drip or flow, and it stays hard when you touch it. [cite: 1]

2. Liquids (The Flowy Ones)

Liquids don't have their own shape. They take the shape of whatever they are in. **Mango Lassi** or **Coconut Oil** are liquids. They flow, drip, and pour easily. [cite: 1]

[Visual: Comparison diagram of a Steel Glass (Solid) and flowing Mango Lassi (Liquid)]

Oobleck: The Rule Breaker!

Oobleck is a special substance that doesn't like following the normal rules of solids and liquids. It is a **Non-Newtonian Fluid**. [cite: 1]

The Mood-Changing Fluid: Oobleck changes its "thickness" based on how you treat it. If you are rough with it, it acts like a solid. If you are gentle, it acts like a liquid! [cite: 1]

The Pressure Secret

- **High Pressure:** When you press hard or punch it quickly, it feels firm like a **Cricket Bat** or a brick! [cite: 1]
- **Low Pressure:** When you let it sit softly in your palm, it oozes and drips like **warm honey** or oil! [cite: 1]

The Oobleck Lab: Making Magic Mud

You can make this amazing substance right in your kitchen! [cite: 1]

Ingredients:

- 2 Cups of **Cornflour** (Makai ka atta). [cite: 1]
- 1 Cup of **Water**. [cite: 1]
- Optional: A few drops of your favorite food colour! [cite: 1]

How to Mix:

1. Pour the cornflour into a large bowl. [cite: 1]
2. Slowly add the water while mixing with your hands. [cite: 1]
3. Keep mixing until it feels like a thick paste that "tears" when you pull it but "melts" when you stop. [cite: 1]

Experiment: The Pressure Test

Now that you've made your Oobleck, it's time to test its magic powers! [cite: 1]

Exhibit 1: The Punch Test

Make a fist and punch the Oobleck surface quickly. [cite: 1]

What happens? Your fist won't sink! It will bounce off as if you hit a solid wall. [cite: 1]

Exhibit 2: The Ooze Test

Now, gently place your open hand on top of the Oobleck. [cite: 1]

What happens? Your hand will slowly sink deep into the goo, just like it's in water! [cite: 1]

[Visual: Two frames showing a fist bouncing off Oobleck and a hand sinking in]

The Science Secret

Why does Oobleck behave so strangely? It's all about how the tiny pieces of cornflour interact with water. [cite: 1]

The "Tangled" Particles

Inside Oobleck, the cornflour bits are like tiny grains of sand. When you **push fast**, the water gets squeezed out from between them, and the grains "lock" together like a solid. When you **go slow**, the water can flow around the grains, allowing them to slide past each other like a liquid! [cite: 1]

SCIENCE ARTIST IDEA

Try making a "ball" of Oobleck by rolling it quickly between your palms. As soon as you stop rolling, watch it "melt" back into a puddle in your hand! [cite: 1]

The Mad Scientist's Clean-up Guide

Science is fun, but a good scientist always cleans up their lab! [cite: 1]

IMPORTANT: PROTECT YOUR PIPES!

NEVER pour Oobleck down the kitchen or bathroom sink. As it settles, it can turn solid and clog the pipes completely! [cite: 1]

How to Clean Up:

- **The Bin:** Scrape your Oobleck into the trash or a compost bin. [cite: 1]
- **Dry & Wipe:** Let any spills on the table dry out. They will turn back into white powder that you can easily wipe or vacuum up. [cite: 1]
- **Hand Wash:** Rinse your hands over a bowl of water first, then pour that water outside or in the trash before using the sink for a final wash. [cite: 1]

Interactive: Solid, Liquid, or Both?

Check the box for each item based on its state of matter! [cite: 1]

Item	Solid	Liquid	Both!
Steel Ladle	☐	☐	☐
Water	☐	☐	☐
Oobleck	☐	☐	☐
Cricket Ball	☐	☐	☐

Lesson Recap

You have successfully solved the Solid-Liquid Mystery! Let's remember our discoveries: [cite: 1]

- **Non-Newtonian:** Oobleck is a fluid that doesn't follow normal rules. [cite: 1]
- **Pressure:** It turns **hard** when you hit it and **runny** when you hold it. [cite: 1]
- **Ingredients:** You only need cornflour and water to make it! [cite: 1]
- **Safety:** Always throw Oobleck in the trash, never down the sink! [cite: 1]

Stay Messy, Stay Curious!

Science is about exploring things that break the rules. See you in the next lab! [cite: 1]